R13

Code No: 117EE

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech IV Year I Semester Examinations, June/July - 2024
LINUX PROGRAMMING
(Computer Science and Engineering)

Time: 3 Hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

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PART – A

(25 Marks)

1.a) What is an associative array? [2]
b) Which command is used to modify the file permissions and access modes in linux? Provide examples. [3]
c) Name the command used to create a directory in linux and provide an example. [2]
d) State the differences between fchown( ) and lchown( ) methods and their usage in linux. [3]
e) How does a process differ from a thread? [2]
f) Why and how exit( ) method is used in Linux? [3]
g) What are the advantages of using Pipes for IPC in Linux? [2]
h) What does the system( ) function do in Linux? Provide its syntax and example. [3]
i) What is Shared Memory in Linux? [2]
j) Mention the differences between stream sockets and datagram sockets. [3]

PART – B

(50 Marks)

2.a) Describe the concept of functions in bash scripting and how they can be defined and invoked.
b) Explain various process utilities available in Linux briefly. [5+5]

OR

3.a) Explain how ‘awk’ can be used in combination with shell scripts and other system utilities to perform complex text processing tasks.
b) Describe the function of the ‘ping’ command in Linux, and how it can be used to test network connectivity. [5+5]

4.a) Compare and contrast ‘open()’ and ‘creat()’ system calls.
b) How does read() system call work in file I/O operations? Explain with a sample code.[4+6]

OR

5.a) Explain the purpose of ‘fstat()’ system call in file operations with an illustrative example code.
b) How do file locking and record locking mechanisms ensure data integrity and security? Explain. [5+5]

6.a) Explore the significance of environment variables in Linux. How do they influence the behavior of programs and processes?

b) What is the purpose of the ‘exec ( )’ family of functions? Explain briefly. [5+5]

OR

7.a) Define the terms "zombie process" and "orphan process" in Linux. What are the causes of these phenomena, and how can they impact system performance?

b) Explain the process of signal generation and handling in Linux. [5+5]

8.a) How do pipes in Linux facilitate communication between processes? Mention their characteristics in terms of data flow and directionality.

b) Explain the concept of implementing IPC mechanism between processes using fifos in Linux. [5+5]

OR

9.a) What distinguishes Message Queues from other IPC mechanisms like pipes and FIFOs?

b) Discuss the kernel support for Message Queues to support IPC in Linux. [4+6]

10.a) What is the role of shared memory in modern multi-threaded and multi-core systems?

b) Provide an example for implementing synchronization between producer-consumer processes using shared memory APIs in Linux. [4+6]

OR

11.a) What are the differences between Unix domain sockets (AF_UNIX) and Internet domain sockets (AF_INET)?

b) Discuss the mechanisms used for binding sockets to specific addresses and ports using the bind() system call in Linux. [5+5]

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